

Time: 3 hours

Marks: 70

[Answer any five questions]

1. a) Suppose $N = \{1, 2, 3, \dots\}$, the positive integers, is the universal set. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$, $C = \{6, 7, 8, 9\}$ and $E = \{2, 4, 6, 8, \dots\}$, the even integer. Then answer the followings: i) $A \cap B = ?$, ii) $B \setminus A = ?$, iii) $(A \cup C)^c = ?$. 3
 b) Draw a Venn diagram of sets A, B, C where A is subset of B, sets B and C are disjoint, but A and C have elements in common. 3
 c) Explain the Inclusion-Exclusion Principle. Show the Theorem for any number of sets that hold the principle. 4
 d) In a class of 80 students, 50 students know English, 55 know French and 46 know German language. 37 students know English and French, 28 know French and German, 7 know none of the languages. How many students know all the 3 languages? 4
2. a) What is recursion? A function f is defined recursively by $f(0) = 3$, $f(n+1) = 2f(n) + 3$. Find the value of $f(4)$. 3
 b) What is recurrence relation? Obtain the recurrence relation for:
 $G(k) = 2 \cdot 4^k - 5 \cdot (-3)^k$ 6
 c) Find the solution to the recurrence relation: $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions $a_0 = 2$, $a_1 = 5$ and $a_2 = 15$. 5
3. a) Define tautology and contradiction. Verify that the proposition $p \vee \neg(p \wedge q)$ is a tautology. 3
 b) Show that the propositions $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are logically equivalent. 3
 c) Consider the conditional proposition $p \rightarrow q$. Construct a truth table for biconditional, converse, inverse and contrapositive of $p \rightarrow q$. 4
 d) Simplify the proposition using laws of algebra of propositions $\neg(p \vee q) \vee (\neg p \wedge q)$. 4
4. a) Define reflexive, symmetric and transitive relation. 3
 b) Consider the following five relations on the set $A = \{1, 2, 3, 4\}$:
 $R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$
 $S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 $T = \{(1, 3), (2, 1)\}$
 $U = A \times A$, the universal relation
 $V = \emptyset$, the empty relation
 Determine whether or not each of the above relations on A is i) Reflexive, ii) Symmetric, and iii) Transitive 6
 c) Define Equivalence relation with example. 2
 d) Let $A = \{1, 2, 3, 4, 5, 6\}$ and let R be the relation on the set A; $R = \{(x, y) \mid x \text{ divides } y\}$. Write R as a set of ordered pairs. 3
5. a) Define Injective, Surjective and Bijective function with example. 3
 b) Give an explicit formula for a function from the set of natural numbers (N) to the set of natural numbers(N) that is 6
 i) one-to-one, but not onto.
 ii) onto but not one-to-one.
 iii) one-to-one and onto.
 iv) neither one-to-one nor onto.
- c) f, g, h are functions on $X = \{1, 2, 3\}$ as 5
 $f = \{(1, 2), (2, 3), (3, 1)\}$
 $g = \{(1, 2), (2, 1), (3, 3)\}$
 $h = \{(1, 1), (2, 2), (3, 1)\}$
 Compute i) $f \circ g \circ h$, ii) $f \circ h \circ g$
6. a) Define with figure: 6
 i) Graph
 ii) Subgraph
 iii) Bipartite graph
 iv) Complete graph
 b) Draw the graph G corresponding to the following adjacency matrix. 3

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$